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Matters arising

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Reply to: Assessment of 3D MINFLUX data for quantitative structural biology in cells

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Check for updates

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REPLYING TO: Prakash, K. & Curd, A. P. *Nat. Methods* https://doi.org/10.1038/s41592-022-01694-x (2022).

Prakash and Curd provide a re-analysis[1] of individual datasets taken from our report[2] demonstrating MINFLUX three-dimensional (3D) imaging in cells. Their evaluation confirms the unique MINFLUX[2,3] localization precision of $\sigma = 1–3$ nm standard deviation. We appreciate their confirmation and also welcome the opportunity to clarify their additional points.

As we have explained in the paper[2], a fluorescence microscope can image only the distribution of fluorophores in the sample—not the labeled biomolecules per se. Therefore, the concept of resolution can only be applied to the fluorophores. As long as fluorophores are individually identified, the localization precision is tantamount to the resolution. By reaching σ on the size scale of the fluorophores themselves (1–3 nm), our paper shows that such remarkable imaging performance can indeed be reached in cells.

Microscopy images from many individual nuclear pore complexes (NPCs) inherently reflect the biological variability of the NPC in cellulo[4], including deviations from roundness and eightfold symmetry[5]. Results gained from averaging over up to a thousand NPCs, as in cryo-electron tomography (cryo-ET), cannot represent the ground truth for each and every NPC in the cell. In fact, NPC images are expected to vary in terms of pore diameter and protein positions, because their conformation is affected by nucleocytoplasmic transport state and forces in the membrane[6–8]. The nanometer-level variations in the MINFLUX data of individual NPCs relative to the averaged cryo-ET rendition actually highlight the ability of this nanoscopy method to access the structural heterogeneity of individual protein complexes. In our view this is a strength and not a weakness of the method, as Prakash and Curd imply.

Since fluorescence microscopy renders just the fluorophores, it is obvious that the degree and type of labeling, the biomolecule–fluorophore distance, and the on/off-switching properties have major consequences for what can be biologically inferred. These factors, however, also depend on the mounting medium, fixation, chemical environment and the type of fluorophore employed in the sample[9]. Moreover, as these factors may vary from sample to sample, drawing general conclusions on an imaging method from individual recordings is questionable. Here we respond to the criticisms raised by Prakash and Curd:

1. "The eightfold symmetry of NPCs is rarely visible at a single nuclear pore level and was not clearly determined in structure-based modeling of the localization datasets". We used the formulation to indicate that the individual corners of the well-known eightfold symmetrical NPC can be identified in the MINFLUX images in general. In fact, owing to incomplete labeling, a variable number of displayed subunits is expected for each pore. The symmetry is further obscured for many of the two-dimensional images by frequent NPC tilts, as in many other super-resolution reports. To account for these tilts, 3D imaging becomes essential. Certainly, camera-based PALM/STORM provides larger fields of view than the scanning-beam MINFLUX demonstrations to date, which is why PALM/STORM can yield many more NPC images in one shot, albeit at lower localization precision. The higher numbers of pores in a field of view facilitate the generation of well-averaged images or the picking of examples that are more completely labeled. Yet we agree with Prakash and Curd that the apparent labeling efficiency in the 3D MINFLUX NPC imaging—not the resolution—was lower than in the referenced STORM study. Among the obvious remedies are labeling optimizations such as replacing the utilized STORM dye Alexa 647, which is not optimal for MINFLUX, with MINFLUX-optimized dyes that are currently under development. The lower apparent labeling is more related to the labeling or the fluorophore used than to the MINFLUX concept.
2. "The mean or best-fit Nup96 ring diameter varies between datasets and the spread of diameters in each dataset is broader than that found by [PALM/STORM]." Even if we consider that fixation may add to the variability of the many individual snapshots of NPCs in various samples, the average diameters (107, 108 and 111 nm) determined by Prakash and Curd still fall within

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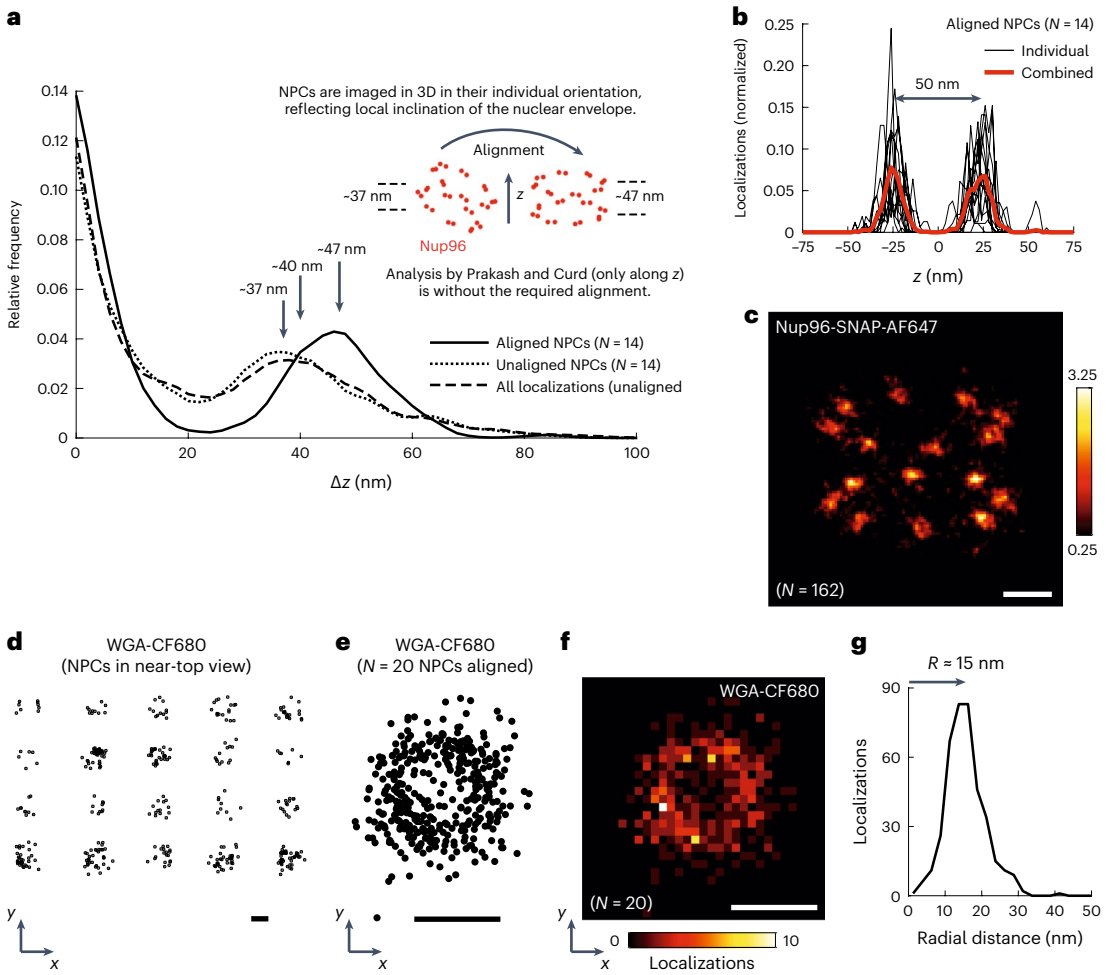


Fig. 1 | Prakash and Curd neglected the tilts of individual NPCs in non-flat nuclear membranes. a–**c**, An analysis on the basis of the RPD along the *z*-axis is inadequate for extraction of the inter-layer distance (ring-to-ring spacing) from Nup96 3D localizations, unless the data from individual NPCs is aligned to fall along the direction of analysis. The individual NPC tilt has to be corrected for to enable accurate measurements. **a**, We estimated the tilts of a subset ($N = 14$) of NPCs for which the labeling was sufficient such that the orientation could be inferred by a 3D fitting analysis. The RPD(Δz) for the unaligned NPCs peaks at ~37 nm (dotted line), reflecting the presence of non-negligible tilts. After their re-orientation to the horizontal, the resulting RPD(Δz) of the NPCs peaks at ~47 nm (bold line), which is close to the ~50 nm expected from cryo-electron tomography. **b**, Histograms of Nup96 localizations from the aligned NPCs along the *z*-direction. Black, individual NPCs; red, 14 NPCs combined. **c**, Averaging result by particle fusion of Nup96 data from 3D MINFLUX, using the code from ref. [10] with the prior knowledge of eightfold pore symmetry. AF, Alexa Fluor 647. Scale bar, 25 nm. **d**–**g**, MINFLUX data of WGA-CF680 binding patterns exhibit a ring-like structure. The individual 3D tilts of NPCs obscure the channel in the majority of NPCs. In top views (x–y), a hollow localization distribution was therefore observed in a small subset of NPCs imaged in chemically fixed U-2 OS cells. **d**, To illustrate the effect, $N = 20$ NPCs, which exhibited a top view or near-top view, were selected by visual inspection. **e**, An overlay of the aligned data forms a ring of ~30 nm diameter. **f**, Histogram of the aligned localizations. **g**, Radial distribution of localizations, indicating a radius of ~15 nm. **d**–**f**, Scale bars, 25 nm.

the respective error margins and are fully compatible with the 107 nm found by cryo-ET after extensive averaging[11]. Therefore, although they are based on small datasets ($N = 20$ in each case), the diameter measurements actually agree very well.

3. "The average *z* distance between cyto- and nucleoplasmic layers of Nup96 localizations was 40.5 nm instead of ~50 nm, in the dataset on which this claim was based". Unfortunately, in their analysis of our data it appears that Prakash and Curd neglected to account for the local inclination of the nuclear envelope (Figure 3f,g of Gwosch et al.[2]). Calculating a relative position distribution (RPD) along the *z*-axis over the entire field of view (see their Figure 2g–j) cannot yield the correct distance between the two Nup96 rings. It biases the RPD towards lower values. We demonstrate this effect by repeating their analysis (Fig. 1a). If the membrane is assumed to be flat and the nuclear pore tilts are not accounted for—that is, the 3D dataset of Figure 3f of Gwosch et al.[2] is used as is—the RPD(Δz) indeed peaks at the erroneous value of ~40 nm (Fig. 1a) derived by Prakash and Curd. Taking the inclinations of individual NPCs into account shifts the RPD peak to a distance of ~47 nm, close to the expected ~50 nm. Histograms of localizations from the aligned NPCs along the *z*-direction (Fig. 1b) exhibit ~50 nm distance between the two layers, as do particle-averaged[10] data from a separate newly acquired recording of a larger number of NPCs (Fig. 1c).
4. "Furthermore, in two-color imaging, the inner ring found in similar [PALM/STORM] experiments at 40-nm diameter[12,13] was not resolved as a ring by MINFLUX". In the STORM reports[12,13] highlighted by Prakash and Curd, rare NPC images were picked by the authors out of many hundreds where a ring is not visible either. Reasons for that may be insufficient labeling, tilts and structural variabilities[7,8]. The visibility of the channel (ring) requires a good top view, which was easier to accomplish in the

referenced STORM study[12] because that study used isolated *Xenopus* nuclei instead of complete mammalian cells, as we did. Nonetheless, the ring-like structure mapped out by WGA can also be extracted from more recently recorded MINFLUX data (Fig. 1d–g) for selected top (x–y) views. An overlay of the aligned MINFLUX data showed a narrowly defined WGA ring of ~30 nm diameter in fixed U-2 OS cells. Yet, care should be exerted when taking these 'rings' for granted. Considering the overall distribution and accessibility of the putative WGA binding partners, further research will be needed to answer if the observed rings are biologically relevant, or whether they are due to labeling and fixation.

5. "[Live MINFLUX and filtering.] We reproduced the published images of Gwosch et al. (2020) and noted an increased level of filtering". The procedure separating emission events from background was performed without the use of prior information about the imaged structures and following clearly described criteria[2] on the basis of the so-called central donut fraction, the estimated location of the molecule with respect to the center of the scan pattern, and the photon number in the last MINFLUX iteration. Bearing in mind the pioneering nature of our experiments, much more data was saved at the time (as a precaution) than eventually had to be used. This naturally led to a large fraction of excluded events in post-processing. More recently published MINFLUX imaging introduced convenient elements of real-time event classification already at the stage of data acquisition[14]. Finally we note that the live-cell MINFLUX images of Nup96 fused to the fluorescent protein mMaple (Figure 2f in Gwosch et al.[2]), outperform similar attempts by PALM.

The choice of fluorophore (and tag) is decisive for the performance of any super-resolution method. Take the cyanine dyes as an example. The strong reversible photoblinking of Cy5 on its own, without the need for a Cy3 activator for switching, was introduced for single-molecule localization microscopy in 2007 by Bock et al.[15] before the effect and its application were corroborated and extended (as the method called dSTORM[16]). The utilized dye Alexa Fluor 647, a close Cy5 relative, has been the go-to STORM dye for one and a half decades. However, owing to key aspects of its photophysical behavior, it is certainly not the optimal dye for MINFLUX. Therefore, comparing MINFLUX with STORM without highlighting the limits of Alexa Fluor 647 as a caveat is neither objective nor definitive.

Molecule-specific imaging at molecule-scale 3D resolution is bound to empower biological research. As for any new fluorescence imaging concept, for MINFLUX we expect continuing progress, especially in conjunction with further advances in fluorescence labeling. Multiple MINFLUX setups are operated in research laboratories around the world, including the new Imaging Centre at the European Molecular Biology Laboratory, which provides open user access to the technology (https://www.embl.org/about/info/imaging-centre/). Supplementary Fig. 1 shows an example of MINFLUX NPC data as it is now routinely obtained by a commercial system. What is even more exciting in our view, is the fact that the hitherto almost inconceivable 3D localization precision of 1–3 nm attained by MINFLUX (and confirmed by Prakash and Curd) is likely to hold the key to an all-optical dynamical structural biology.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-022-01695-w.

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Methods

The MINFLUX data shown reanalyzed in Fig. 1a,b was taken from Figure 3f of Gwosch et al.[2], and the data presented in Fig. 1c,d–g was recorded anew on an Abberior Instruments MINFLUX nanoscope[14] following previously established fluorescence labeling and imaging protocols[13,14].

Cell culture

Before seeding of cells, high-precision 18-mm round glass coverslips (No. 1.5H, Marienfeld) were cleaned by placing them in ethanol overnight, rinsed with water and dried in a laminar flow cell culture hood before finally irradiating the coverslips with ultraviolet light for 30 min.

Nup96-SNAP cells (300444, CLS) were seeded on clean glass coverslips 2 days before fixation to reach a confluency of about 50–70% on the day of fixation. The cells were grown in growth medium (Dulbecco's modified Eagles's medium (DMEM; 11880-02, Gibco)) containing 1× MEM NEAA (11140-035, Gibco), 1× GlutaMAX (35050-038, Gibco) and 10 % (v/v) fetal bovine serum (10270-106, Gibco) at 37 °C and 5% CO_2. Before further processing, the growth medium was aspirated, and samples were rinsed with PBS (room temperature) to remove dead cells and debris.

Preparation of NPC samples

Cells on glass coverslips were prefixed in 2.4% (w/v) formaldehyde (15710, EMS) in PBS for 20 s before incubating them for 10 min in 0.5% (v/v) Triton X-100 in PBS. Fixation was completed in 2.4 % (w/v) formaldehyde in PBS for 20 min. Formaldehyde was quenched for 5 min in 100 mM NH_4Cl in PBS and then washed three times for 5 min in PBS. Fixed cells were blocked with Image-IT signal enhancer (Thermo Fisher) for 30 min and then incubated with 1 µM BG-AF647, 0.5 % bovine serum albumin and 1 mM dithiothreitol in PBS for 1 h to stain Nup96-SNAP-tag. Cells were washed three times for 5 min with PBS and subsequently imaged.

For imaging with WGA-CF680 the sample was incubated for 10 min with 1:10,000 diluted WGA-CF680 (29029-1, Biotium) in 100 mM Tris, pH 8.0, 40 mM NaCl, and rinsed three times with PBS shortly before mounting. Samples were mounted on glass slides (12290, BMS Microscopes) with a small cavity to hold the imaging buffer and sealed with 2K silicone (eco-sil, 1300 9100, Picodent).

Imaging buffer

Glucose oxidase/catalase buffer supplemented with cysteamine (MEA) was used to image Nup96-SNAP-AF647 and WGA-CF680 samples. GLOX + MEA contained 50 mM Tris/HCl pH 8, 10 mM NaCl, 10 % (w/v) D-glucose, 500 µg ml^{-1} glucose oxidase, 40 µg ml^{-1} catalase and 14 mM MEA in H_2O (all reagents from Sigma).

Data analysis

The described analysis was performed in MATLAB. In brief, for Fig. 1a the RPD(Δz) was calculated for three separate sets of 3D localization coordinates: (1) all localizations within the field of view, unaligned; (2) 14 segmented NPCs, unaligned; and (3) the same 14 segmented NPCs after horizontal alignment (rigid transformation of their coordinates), removing the tilts estimated from a comparison to a template of the average positions of Nup96 C termini on the basis of the cryo-ET structure[11]. For the localization coordinate set (1), all Δz across the entire field of view were calculated, but adding a filter in the Euclidian distance $R < 200$ nm for meaningful interpretation of the RPD histogram, owing to the substantial non-flatness of the nuclear membrane over the field of view. Figure 1b shows direct, normalized z histograms of the aligned localizations of set (3). For Fig. 1c, 162 NPCs were segmented from the field of view of a Nup96 recording by visual inspection, and their localization data was averaged using the code of ref. [10] with the prior knowledge of eightfold pore symmetry. The data are displayed as a sum projection (smoothed histogram; isolated localizations at density <1/1,000 nm^3 removed). The combined set of WGA-CF680 x,y localizations from 20 NPCs (localizations in Fig. 1e, histogram in Fig. 1f) was obtained by fitting circles to the individual localization patterns of Fig. 1d, followed by co-alignment on the basis of their centers.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper.

Code availability

The analysis code is available in Supplementary Software.

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Author contributions

The reply was written by K.C.G., F.B., J.K.P., P.H., J.E., J.R., U.M., R.S., S.J.S. and S.W.H. F.B. and S.J.S. performed data analysis. U.M. and R.S. recorded MINFLUX data.

Competing interests

S.W.H. is a co-founder of the company Abberior Instruments, which commercializes super-resolution microscopy systems, including MINFLUX. The Max Planck Society holds patents on the principles, embodiments and procedures of MINFLUX, benefiting S.W.H., K.C.G. and F.B. U.M. and R.S. are employees of Abberior Instruments GmbH.

Additional information

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Software and code

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Data collection	Abberior Instruments Imspector software with MINFLUX drivers
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